

Does Higher Socioeconomic Status Associate with Better Mental Health Outcomes Among Primary Caregivers in Rural China?

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Abstract:

This study explores the intricate relationship between socioeconomic status (SES) and the mental health outcomes of primary caregivers in rural China, focusing on the Early Childhood Development (ECD) context in Ya'an. Anchored in a major intervention project launched in 2023, the study critically examines the hypothesis that higher socioeconomic status correlates with lower levels of negative emotions among caregivers.

Utilizing cross-sectional data, the research investigates how educational attainment and household income influence caregivers' mental health, measured through the DASS-21 score in three dimensions: stress, anxiety, and depression. The findings reveal that higher household income and educational levels are significantly associated with reduced negative emotions, underscoring the protective effects of financial stability and education. Additionally, the study highlights distinct mental health challenges among different demographic groups, particularly emphasizing the heightened vulnerability of elderly caregivers and females.

The results advocate for targeted mental health interventions and policies that address both immediate symptoms and broader socioeconomic determinants. Investment in education and healthcare is identified as crucial for improving caregivers' well-being and fostering a supportive environment for early childhood development. This study also calls for future research to investigate the role of parents' mental well-being as a mediator between socioeconomic status and children's cognitive development, offering deeper insights into how poverty impacts children's long-term well-being. Additionally, exploring mental health disparities between genders can help understand the unique challenges faced by each group, informing the development of gender-specific mental health interventions. Incorporating subjective indicators such as Subjective Social Status (SSS) would also be valuable in examining the dynamics of perceived social positions and their impact on mental health. By addressing these areas, the study aims to inform policy and programmatic strategies that enhance the mental health and well-being of caregivers and their children in rural settings.

1. Introduction

In the confluence of psychology, public health, and social sciences, the intricate relationship between socioeconomic status (SES) and mental health outcomes has increasingly become a pivotal area of inquiry. This relationship is particularly salient among caregivers, a group noted for their vulnerability to stress and mental health challenges due to the profound responsibilities inherent in their roles. Focusing on Early Childhood Development (ECD) within the rural settings of Ya'an, China, this study is anchored in a major intervention project launched in 2023. It critically assesses the prevailing assumption that higher socioeconomic status correlates with diminished negative emotions among caregivers.

By delving into the complex dynamics between socioeconomic factors—specifically education and income—and mental health outcomes, this research explores the nuanced effects of SES on caregiver mental health in a rural Chinese context. The study's findings aim to inform future policy and programmatic strategies by providing valuable insights into the socio-economic underpinnings of mental health among rural caregivers. Ultimately, this research seeks to enhance our understanding of how to effectively support these caregivers, thereby fostering a nurturing environment conducive to the optimal early development of children in rural areas.

2. Literature Review

The foundational study by Adler et al. (1994) represents a cornerstone in understanding the SES-health gradient, illustrating that health disparities extend across the entire socioeconomic spectrum. This study reveals that even slight differences in SES can lead to significant variations in health outcomes including mental health. This insight is particularly relevant when examining the experiences of caregivers. Given their extensive responsibilities, caregivers are inherently more susceptible to stress and mental health challenges. This suggests that caregivers from different socioeconomic backgrounds may face differing degrees of mental health issues related to their socioeconomic status.

Building upon this foundational understanding, Gallo & Matthews (1999) explore the mechanisms behind the SES-health gradient, proposing that stress and anxiety associated with lower SES could exacerbate mental health problems through the mediation of negative emotions. This concept is of particular relevance to caregivers, who frequently face a confluence of emotional and financial stressors. It indicates that the mental health repercussions of caregiving might be intensified for those from lower socioeconomic backgrounds.

Further emphasizing the mental health challenges faced by caregivers, Bin Sallim et al. (2015) shed light on the high prevalence of mental health disorders among caregivers of Alzheimer's patients. Although SES factors were not explicitly measured in their study, the implications of SES in shaping these mental health outcomes suggests a significant area for further research. Complementing this findings, Oshio (2014) reports that in Japan, involvement in caregiving is negatively associated with mental health, with SES playing a contributing factor to this adverse relationship. Collectively, these findings underline that the mental health of caregivers is influenced not only by their caregiving responsibilities but also significantly by their socioeconomic status.

Deepen the understanding, the work of Muñoz-Bermejo et al. (2020) explores the crucial role of perceived social support among elderly caregivers, revealing a direct correlation with better mental health outcomes. This finding highlights the protective effect of social support, suggesting it could alleviate some of the mental health challenges inherent in caregiving. Particularly, this insight posits that social support not only cushions the caregiving stressors but also may potentially mitigate the adverse effects tied to lower socioeconomic status. It implies that the strength of one's social network could be as critical to a caregiver's mental health as their material resources, highlighting an essential area for further investigation and intervention.

Building on this nuanced findings, Demakakos et al. (2008) and Adler et al. (2000) shift the focus to the concept of Subjective Social Status (SSS). Their investigations reveals an intriguing aspect: the perception of one's social standing often has a more significant impact on health than the actual, objective socioeconomic conditions. These perceptions, interpreted through the lens of societal and cultural norms, may trigger stress responses or feelings of social comparison. Such effects are notably pronounced in diverse societal contexts - from the collectivist frameworks prevalent in China to the individualistic ethos of the United States. This insights emphasizes the complex relationships between subjective social perceptions and health outcomes. It underscores the need for a deeper understanding of how societal contexts may influence mental health.

Early Childhood Development (ECD)

The early years of life, particularly the first three, are increasingly recognized as a pivotal phase for Early Childhood Development (ECD). This period, as detailed by Black et al. (2017), is marked by significant brain development, because it lays the foundation for future skill development and life outcomes. A stimulating environment during these early years is crucial especially for fostering cognitive growth, linguistic abilities, and long-term health, establishing a direct link between early environmental influences and developmental trajectories ("Improving Early Childhood Development," n.d.). Recognizing the profound impact of these early years, Schady et al. (2015) and Shawar & Shiffman (2017) argue that investing in the development of disadvantaged children not only enhances human capital but also acts as a strategic intervention to prevent the perpetuation of poverty across generations. In response, global organizations such as the World Bank, the Inter-American Development Bank, the United Nations, and the World Health Organization have prioritized ECD initiatives aiming to enrich the developmental environments of young children worldwide (Black et al., 2015).

Parallel to the emphasis on ECD, is a growing awareness of the critical role caregivers' mental health plays in shaping these early experiences. Research indicates that maternal mental health issues are related to significant developmental challenges for children, affecting their cognitive, social, and academic growth later years (Gelaye et al., 2016; Yue et al., 2018; S. Zhang et al., 2018). Although establishing a direct causal relationship between maternal mental health and ECD outcomes is complex, with potential environmental and genetic factors at play (Burke, 2003; C. Zhang et al., 2017), there are evidences suggest that mental health struggles can lead caregivers to engage less in stimulating interactions, such as reading or singing to their children (Yue et al., 2018; Zhong et al., 2021). This observation has propelled calls for interventions to enhance caregivers' mental health, which has now recognized as a crucial facet of global public health efforts (Gelaye et al., 2016).

The concern over caregivers' mental well-being is highlighted by the World Health Organization's recent research, which reports that about 13% of women experience mental health issues postpartum (World Health Organization, 2020). The potential factors for such conditions are multifaceted. It involves socio-economic challenges, lower social status, and a paucity of supportive networks for mothers (Atif et al., 2015; Fisher et al., 2012). This is particularly pronounced in rural settings, where women, especially for those in remote areas from urban centers, exhibit a higher prevalence of mental health issues (Qiu et al., 2016).

Early Childhood Development (ECD) and caregiver's mental health in China

ECD delays and caregiver mental health issues are notably prevalent in rural China. Recent research by Emmers et al. (2021) highlights this concern, estimating that approximately 45% of rural Chinese three-year-olds may exhibit cognitive delays. Moreover, S. Zhang et al. (2018) found that 40% of rural caregivers suffer from major mental health issues, including anxiety, depression, and stress. These conditions not only affect the caregivers but also have been proved link to ECD delays in children, which has established a compelling case for targeted interventions (S. Zhang et al., 2018; Zhong et al., 2021).

Interventions to address these challenges can be categorized based on their approach to caregiver mental health into three groups: those with specific components targeting caregiver mental health, those without targeted mental health components but applying mental health-friendly delivery strategies, and those lacking the both. Some studies focusing on the first category, which specifically target caregiver mental health, often report significant improvements in both caregiver mental well-being and child development outcomes (Lakkis et al., 2020; Singla et al., 2015). Meanwhile, interventions that do not directly target mental health but utilize mental health-friendly strategies have also shown promise in reducing symptoms of maternal depression in three out of four studies examined (Baker-Henningham et al., 2005; Hamadani et al., 2019). While interventions lacking both targeted components and mental health-friendly strategies have produced mixed results; while some have observed improvements in maternal well-being (Aboud et al., 2013; Andrew et al., 2020), the majority have not found significant impacts (Attanasio et al., 2014; Chang et al., 2015; Luoto et al., 2021).

In a recent initiative, we conducted a study in Ya'an to investigate the potential impacts of parenting centers on the mental health of caregivers across approximately 12 rural villages. These centers provide a comprehensive suite of services, including individual parenting training, group activities, and open play spaces for both caregivers and their children. A key focus of our approach is to deliver organized social and emotional support through group activities and the incorporation of positive language during training sessions.

This investigation forms part of broader intervention project deployed in Ya'an, with a specific aim to critically evaluate the underlying assumption that higher socioeconomic status (SES) correlates with improved mental health outcomes, or lower negative emotions, among caregivers. This exploration examines the intricate dynamics between socioeconomic factors—specifically education and income—and mental health outcomes measured by DASS-21 score (Lovibond, S.H.,

& Lovibond, P.F.,1995) among caregivers in rural areas. By analyzing the effects of SES on caregiver mental health within the context of Ya'an, the study seeks to provide insights that could inform future policy and programmatic efforts. Ultimately, it aims to enhance our understanding of how best to support caregivers in this region, thereby ensuring a more nurturing and conducive environment for the early development of children in Chinese rural settings.

3. Sample Choosing Strategy

Our sample was selected through a randomized controlled trial (RCT) conducted across various villages, premised on the hypothesis that higher socioeconomic status correlates with better mental health outcomes for caregivers. The study encompasses a dataset consisting of 922 observations related to income and 919 observations concerning education from Ya'an, Sichuan Province, China, as of 2023.

The ethnic makeup of the targeted sample area is predominantly Han, which is reflective of China's majority ethnic group. This region is known for having a significant number of impoverished counties nationally, making it a representative sample of typical rural areas. Such selection criteria are critical in minimizing the influence of potential exogenous factors, such as specific policies targeting ethnic minorities or varying levels of regional development, which could skew the research outcomes. Additionally, these areas are expected to house an adequate number of parenting centers, thereby enhancing the relevance and feasibility of the study.

To ensure a robust sample size, only townships with populations exceeding six hundred individuals were considered. Moreover, with a focus on early childhood development, the study specifically targeted families with newborn children aged 6 to 36 months. Initially, 12 villages from 5 counties were selected. Sample families and their caregivers were chosen based on their overall socioeconomic status (SES), assessed through criteria such as family income, education level, and occupation, with assistance from local family planning officials. The most recent baseline data, collected in 2023, comprises 1,100 caregivers from the rural areas of Ya'an, providing a comprehensive overview of the caregiving environment in these communities.

4. Research Model Specification

a) Research Model

To facilitate a more intuitive analysis of the relationship between socioeconomic status and mental health outcomes among primary caregivers in Ya'an, our model is structured as follows: Socioeconomic status is quantified through two key indicators—household income and the educational level of primary caregivers. The mental health outcomes are measured by the Depression, Anxiety, and Stress Scale (DASS-21), which assesses the emotional states of depression, anxiety, and stress. This model followed is designed to examine how variations in socioeconomic factors influence the mental health of caregivers.

(1) Basic Model

Income

$$\text{dass_total}_{it} = \alpha_0 + \alpha_1 \text{income}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_stress}_{it} = \alpha_0 + \alpha_1 \text{income}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_anxiety}_{it} = \alpha_0 + \alpha_1 \text{income}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_depression}_{it} = \alpha_0 + \alpha_1 \text{income}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

Education

$$\text{dass_total}_{it} = \alpha_0 + \alpha_1 \text{edu}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_stress}_{it} = \alpha_0 + \alpha_1 \text{edu}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_anxiety}_{it} = \alpha_0 + \alpha_1 \text{edu}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

$$\text{dass_depression}_{it} = \alpha_0 + \alpha_1 \text{edu}_{it} + \alpha_3 \text{control}_{it} + \delta_i + \rho_t + \varepsilon_{it}$$

Where i indexes individuals, t indexes time. Both the logarithm of household income (income) and the primary caregiver's education level (edu) serve as the core explanatory variables. The mental health outcome (dass_total), stress (dass_stress), anxiety (dass_anxiety), and depression (dass_depression) are the dependent variables. The *control* represents a set of relevant control variables, including internet access (inter), air conditioning access (air_conditioning), the presence of doctors in the villages (doctor), households with car ownership (car), age of the primary caregivers (age), the self-reported health status of the primary caregivers (health), the child's gender (c_gender), and the caregivers's expectation for child's educational (c_study); δ_i represent the individual fixed effects, capturing time-invariant unobservable heterogeneity, thereby isolating the impact of varying socioeconomic conditions over time; ρ_t denotes the time effects, and ε_{it} is the error term.

b) Descriptive Statistical Results

Table 1

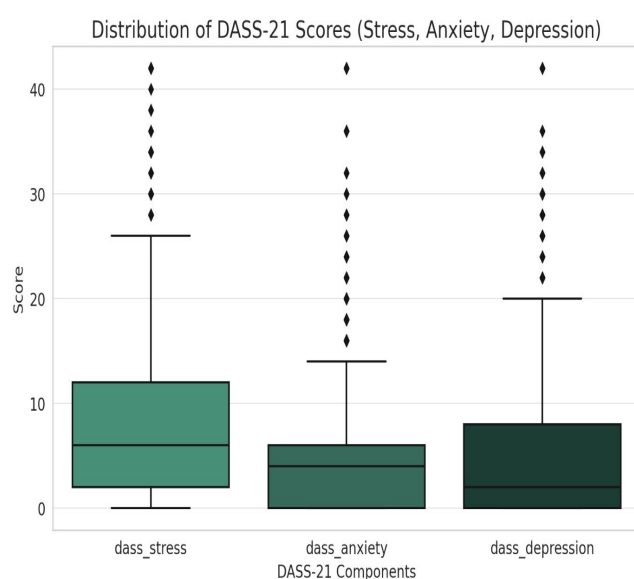
Variables	N	Mean	(Median) p50	SD	Min	Max
dass_total	922	18.985	15.000	17.606	1.000	89.000
log_dass_total	922	2.395	2.708	1.211	0	4.419
dass_stress	922	9.128	7.0	7.6	1.0	43.0

dass_anxiety	922	5.957	5.0	5.9	1.0	43.0
dass_depression	922	6.106	3.0	6.4	1.0	43.0
income	922	54674.3	47501.0	460147.7	1.0	2.50e+05
log_income	922	10.43	10.78	1.624	0	12.43
edu	922	2.433	2	1.081	0	5
internet	922	0.974	1	0.159	0	1
air_conditioning	922	0.623	1	0.485	0	1
doctor	922	0.302	0	0.459	0	1
car	922	0.757	1	0.429	0	1
age	922	33.7	30	10.6	19.0	62.0
log_age	922	3.474	3.401	0.287	2.944	4.094
health	922	0.638	1	0.481	0	1
c_gender	922	0.535	1	0.499	0	1
c_study	922	2.447	3	0.707	1	3

Note: edu: 1= no school attended; 2= primary school; 3= junior high; 4=senior high; 5= college and above;
c_study: 1= moderately anticipated; 2= anticipated; 3= highly anticipated.

An initial statistical analysis was performed to delineate the fundamental characteristics of the dataset, as presented in Table 1 above.

Mental Health Outcomes: The overall level of negative emotions, measured by DASS score ("*log_dass_total*") exhibited a moderate mean score of 2.395 and a median of 2.708. The median exceeding the mean suggests a distribution skewed towards higher scores, indicative of a greater prevalence of negative emotions among some caregivers. The standard deviation of 1.211 points to a broad variability in emotional states across the sample.



The boxplots depict the three demensions of distribution of DASS-21 score - Depression, Anxiety, and Stress Scale among caregivers in rural Ya'an.

Stress: The median indicates moderate stress with outliers suggesting that some caregivers experience significantly higher stress levels.

Anxiety: Features a lower median with a broader range, indicating variability in anxiety levels.

Depression: Shows the lowest median with a narrow range, suggesting that depression is less prevalent.

Socioeconomic Indicators: The logarithm of household income ("*log_income*") ranged from 0 to 12.43, with a mean of 10.43 and a median of 10.78, suggesting a relatively symmetric distribution around the central values. However, the standard deviation of 1.624 indicates considerable income disparity within the sample. The educational level of the primary caregivers ("*edu*") averaged at 2.433, with a median of 2, which is lower than the mean. This skewness towards lower educational attainment highlights educational challenges in the region.

Control Variables - Resource Access and Demographic and Health Variables: Access to the internet ("*internet*") is nearly universal among the sample, with the majority having air conditioning ("*air_conditioning*"). The age of primary caregivers ("*log_age*") appears relatively homogenous with a small standard deviation (0.287). The self-reported health status of primary caregivers ("*health*") indicates that a majority report being healthy. The child's gender ("*c_gender*") is almost evenly distributed; the caregiver's expectation for the child's education ("*c_study*") are moderately high on average — from "1" to "3" means from moderate to high — meaning that most caregivers have significant educational aspiration for their children.

c) Multicollinearity and Correlation Analyses

i. VIF Multicollinearity Test

In our analysis, the Variance Inflation Factor (VIF) values for all included variables were found to be below the commonly accepted threshold, with each value under 10, and with their reciprocal values ($1/VIF$) all greater than 0.1. The Mean VIF was 1.06, also below the threshold of 10, indicating a minimal level of multicollinearity among the variables, as the table 2 shows below:

Table 2

Variables	VIF	1/VIF
log_age	1.14	0.875183
health	1.1	0.911572
car	1.09	0.914668
air_conditioning	1.07	0.938705
internet	1.06	0.944607
c_study	1.03	0.971375
doctor	1.02	0.981243
c_gender	1.01	0.991257
MeanVIF	1.06	

The above table reinforces the conclusion that the control variables within the model are not strongly intercorrelated, thus supporting the reliability of the regression coefficients derived from the dataset, and ensuring that our model's estimations are both stable and trustworthy.

ii. Correlations Analysis

The study performed a correlation analysis on a set of key variables including the logarithm of household income (*log_income*), the educational level of primary caregivers (*edu*), the level of mental health outcome (*log_dass_total*), internet access (*internet*), and the presence of airconditioning (*air_conditioning*), among others. See as the table 3 below.

Table 3

	log_dass_total	log_income	edu	internet	air_conditioning	doctor	car
log_dass_total	1						
log_income	-0.097***	1					
edu	-0.097***	0.163***	1				
internet	-0.070**	0.0400	0.059*	1			
air-conditioning	-0.067**	0.121***	0.103***	0.112***	1		
doctor	-0.0190	0.0180	0.0320	0.0480	0.00500	1	
car	-0.0150	0.156***	0.189***	0.177***	0.211***	0.0310	1
log_age	-0.0380	-0.201***	-0.451***	-0.136***	-0.0110	-0.102***	-0.146***
health	-0.221***	0.165***	0.223***	0.104***	0.083**	0.097***	0.078**
c_gender	0.0500	0.00400	-0.0190	0.0250	0.0500	-0.0460	0.0340
c_study	-0.056*	-0.117***	-0.097***	-0.0510	-0.068**	-0.00100	-0.0140
	log_age	health	c_gender	c_study			
log_age	1						
health	-0.267***	1					
c_gender	0.0480	0.0120	1				
c_study	0.147***	-0.00200	-0.00400	1			

(The symbols “*”, “**”, and “***” indicate statistical significance at the 1%, 5%, and 10% levels, respectively.)

Income, Education and Mental Health: There is both a statistically significant negative correlations of -0.097 between household income (*log_income*) and the level of mental health outcomes (*log_dass_total*) and between primary education level (*edu*) and their mental health level (*log_dass_total*) at the 1% level. This suggests that higher income and education levels are associated with lower levels of negative emotions.

Income and Education: A significant positive correlation of 0.163 at the 1% level between income and the educational level of primary caregivers (*edu*) indicates that higher household income is linked to greater educational attainment among primary caregivers.

Income and Access to Amenities: There are positive correlations between income and amenities such as air conditioning (*air_conditioning*) and car ownership (*car*), significant at the 1% level. These findings demonstrate that households with higher income levels are more likely to possess these amenities.

Education and Amenities: Education also shows a significant positive correlation with the same amenities, indicating that higher educational levels among caregivers enhance the likelihood of accessing better household facilities.

Self-reported Health and Socioeconomic Factors: Self-reported health (*health*) exhibits a significant positive correlation with income and a significant negative correlation with *log_dass_total*, implying that higher income not only correlates with better reported health but also with lower levels of negative emotions. This supports the hypothesis that socioeconomic status, through channels such as improved health, can mitigate negative emotions.

Primary Caregiver's Age and Income and Education: There is a significant negative relationship at the 1% level between the age of primary caregivers and their income and education levels. As the age of primary caregivers increases, their income and education levels tend to decrease. This inverse relationship highlights the economic and educational challenges faced by older caregivers.

Overall, the absence of extremely high or low correlation coefficients (close to -1 or 1) suggests that there is no perfect multicollinearity among the variables, validating the reliability of the data for further regression analysis.

5. Empirical Analysis

a) Baseline Regression

Building upon the previously established indicators, model specifications, and validation efforts, this analysis utilizes an Ordinary Least Squares (OLS) model to conduct empirical evaluation using baseline cross-sectional data.

i. Impact of Household Income on Primary Caregivers' Mental Health Outcome

Table 4

	log_dass_total	log_dass_total
log_income	-0.07232*** (-2.96)	-0.06353*** (-3.69)
internet		-0.46736** (-2.32)
air_conditioning		-0.09969 (-1.22)
doctor		-0.01333 (-0.16)
car		0.05236 (0.55)
log_age		-0.48640*** (-3.27)
health		-0.57411*** (-6.94)
c_gender		0.14431*

		(1.86)
c_study		-0.10200*
		(-1.81)
_cons	3.15534***	5.77348***
	(12.23)	(9.08)
<i>N</i>	919	919
<i>R</i> ²	0.009	0.076

(Note: The figures in parentheses represent standard errors. The symbols "****", "***", and "**" indicate statistical significance at the 1%, 5%, and 10% levels, respectively.)

In our model, household income serves as the independent variable. Column (1) of the results showcases the regression using the logarithm of household income ($\log_income$) alone as the explanatory variable, revealing a negative coefficient of -0.07232, significant at the 1% level. This indicates a robust inverse relationship between household income and negative emotional states. When control variables are introduced in Column (2), the coefficient for household income shifts slightly to -0.06353, maintaining its negative correlation and statistical significance. This consistency suggests that the substantial role that higher income levels play in mitigating negative emotions, likely through enhanced living conditions, improved healthcare, and increased access to educational and recreational opportunities.

The stability of the income coefficient across both models suggests an independent effect of income on emotional well-being that transcends the accounted control variables. This implies that the advantages of higher income extend beyond the immediate material benefits.

Table 5

	log_dass_stress	log_dass_anxiety	log_dass_depression
log_income	-0.03978** (-2.51)	-0.05098*** (-3.23)	-0.05468*** (-2.66)
internet	-0.28693 (-1.47)	-0.37045** (-1.97)	-0.50600*** (-2.61)
air_conditioning	-0.08751 (-1.25)	-0.13322** (-1.96)	-0.06740 (-0.94)
doctor	0.02896 (0.40)	-0.00682 (-0.10)	-0.02152 (-0.28)
car	0.03897 (0.46)	0.05163 (0.64)	0.00370 (0.04)
log_age	-0.37648*** (-2.85)	-0.30344** (-2.51)	-0.47842*** (-3.73)
health	-0.44171*** (-6.16)	-0.46868*** (-6.73)	-0.46433*** (-6.31)
c_gender	0.10269 (1.53)	0.06758 (1.06)	0.07772 (1.15)

c_study	-0.11080** (-2.25)	-0.06865 (-1.47)	-0.09153* (-1.84)
_cons	4.29664*** (7.42)	3.75206*** (7.19)	4.53969*** (7.80)
<i>N</i>	919	919	919
<i>R</i> ²	0.059	0.073	0.070

By further subdividing the measurement of DASS 21 score into specific categories: stress (log_dass_stress), anxiety (log_dass_anxiety), and depression (log_dass_depression), with each category serving as a dependent variable in the regression analysis. The coefficients for household income across all three categories of negative emotions are negative and statistically significant, indicating that higher income levels are consistently associated with lower levels of stress, anxiety, and depression.

Specifically, a higher income is associated with lower stress (coefficient of -0.03978, significant at the 5% level) due to increased financial security that allows for better housing, healthcare, and leisure opportunities. Similarly, anxiety shows a significant negative correlation (coefficient of -0.05098 at the 1% level) as greater income provides access to quality education and career opportunities, promoting personal growth and mitigating educational and occupational anxieties. Additionally, higher income reduces the likelihood of depression (coefficient of -0.05468, significant at the 1% level) by enhancing life satisfaction through improved quality of life.

This observation is further elucidated in both Tables 4 and 5, where an intriguing trend emerges: as the age of primary caregivers increases, their reported happiness also tends to rise. This finding is somewhat counterintuitive and contradicts earlier studies that typically associate increased age with heightened vulnerability to mental health challenges among caregivers. This discrepancy might suggest unique socio-cultural or environmental factors at play in the studied population that mitigate the typical stressors associated with aging. Further investigation into the specific supports, community structures, or coping mechanisms prevalent among older caregivers in this setting could provide valuable insights into this unexpected positive correlation.

ii. Impact of Household Education on Primary Caregivers' Mental Health Outcome

Table 6

	log_dass_total	log_dass_total
edu	-0.10821*** (-2.94)	-0.12746*** (-3.18)
internet		-0.48312** (-2.37)
air_conditioning		-0.10463 (-1.29)

doctor		-0.00935 (-0.11)
car		0.05570 (0.58)
log_age		-0.64202*** (-4.02)
health		-0.57501*** (-6.91)
c_gender		0.14649* (1.90)
c_study		-0.08738 (-1.56)
_cons	2.65862*** (27.17)	5.93444*** (8.98)
<i>N</i>	922	922
<i>R</i> ²	0.009	0.080

Table 6 presents data that elucidate the relationship between the education level of primary caregivers and the mental health of individuals under their care. The results indicate a clear inverse correlation: as the education level of caregivers increases, there is a notable decrease in the negative emotional states experienced by those they care for.

This finding supports the notion that higher educational attainment not only enhances emotional support and material assistance but also correlates with improved communicative abilities, emotional intelligence, and problem-solving skills. These competencies are essential for effective caregiving and may significantly alleviate stressors associated with caregiving roles, thus improving the overall mental health outcomes of both caregivers and those they support. This evidence underscores the value of educational initiatives aimed at caregivers as a means of enhancing their ability to manage caregiving challenges and promote healthier emotional environments.

Table 7

	log_dass_stress	log_dass_anxiety	log_dass_depression
edu	-0.07459** (-2.11)	-0.12019*** (-3.48)	-0.07327** (-2.06)
internet	-0.29543 (-1.53)	-0.38712** (-2.00)	-0.51130*** (-2.63)
air_conditioning	-0.09196 (-1.31)	-0.13274** (-1.97)	-0.07650 (-1.07)
doctor	0.03253 (0.45)	-0.00677 (-0.10)	-0.01749 (-0.23)
car	0.03904	0.06122	-0.00331

	(0.46)	(0.77)	(-0.04)
log_age	-0.46656*** (-3.29)	-0.45208*** (-3.50)	-0.55334*** (-4.02)
health	-0.44451*** (-6.20)	-0.46294*** (-6.68)	-0.46787*** (-6.32)
c_gender	0.10457 (1.56)	0.06805 (1.07)	0.08209 (1.21)
c_study	-0.10070** (-2.06)	-0.05964 (-1.29)	-0.07573 (-1.53)
_cons	4.35817*** (7.35)	4.00900*** (7.36)	4.38011*** (7.54)
<i>N</i>	922	922	922
<i>R</i> ²	0.061	0.081	0.068

As demonstrated in Table 7, the education level of primary caregivers exhibits statistically significant negative associations with all three mental health components analyzed in this study—stress, anxiety, and depression—indicating that higher educational attainment correlates with lower reported levels of these negative emotions.

These findings reinforce the protective role of education in mitigating mental health challenges and support the theory that higher educational levels equip individuals with more effective strategies and resources to manage life's adversities.

Furthermore, the variables 'age' and 'health' stand out as significant predictors across all models, reaffirming the notion that older caregivers and those in better self-reported health experience lower levels of negative emotions. The implications of these findings suggest that mental health interventions should not solely focus on alleviating immediate symptoms but also encompass broader educational and health promotion strategies to fortify individuals against stressors.

b) Heterogeneity Analysis

Table 8

	DASS - Income		
	Dads	Mom	Grandparents
log_income	-0.04238 (-0.91)	-0.03970 (-1.46)	-0.08725*** (-2.90)
internet	-0.74387** (-2.19)	-0.40900 (-1.47)	-0.51148 (-1.56)
air_conditioning	-0.40297* (-1.68)	-0.03191 (-0.33)	-0.21335 (-1.01)
doctor	-0.01587	-0.02042	0.01154

	(-0.07)	(-0.21)	(0.04)
car	0.26915	-0.03423	0.19830
	(1.07)	(-0.29)	(0.88)
log_age	0.19049	-0.49437*	-0.01527
	(0.29)	(-1.69)	(-0.01)
health	-0.55382**	-0.52365***	-0.73511***
	(-2.17)	(-5.29)	(-3.56)
c_gender	0.23664	0.15267*	0.14417
	(0.94)	(1.65)	(0.73)
c_study	-0.13577	-0.10176	-0.02707
	(-0.96)	(-1.55)	(-0.16)
_cons	3.45831	5.48863***	4.01558
	(1.42)	(5.25)	(0.69)
<i>N</i>	89	654	176
<i>R</i> ²	0.151	0.058	0.134

Table 9

DASS - Education			
	Dads	Moms	Grandparents
edu	0.15915	-0.13318***	-0.26369**
	(1.20)	(-2.90)	(-2.34)
internet	-0.47064	-0.45074	-0.41099
	(-1.03)	(-1.62)	(-1.19)
air_conditioning	-0.47579**	-0.02361	-0.19344
	(-2.09)	(-0.24)	(-0.92)
doctor	-0.01274	-0.02431	0.06037
	(-0.06)	(-0.25)	(0.21)
car	0.21215	-0.01185	0.17904
	(0.82)	(-0.10)	(0.81)
log_age	0.46598	-0.58370**	0.31632
	(0.71)	(-2.00)	(0.24)
health	-0.59897**	-0.50970***	-0.71267***
	(-2.31)	(-5.20)	(-3.40)
c_gender	0.30183	0.16364*	0.13344
	(1.24)	(1.78)	(0.66)
c_study	-0.15945	-0.09358	-0.01710
	(-1.14)	(-1.44)	(-0.10)
_cons	1.55232	5.70546***	2.08594
	(0.62)	(5.41)	(0.38)
<i>N</i>	89	656	177
<i>R</i> ²	0.157	0.068	0.139

By categorizing the sample into three groups—Moms, Dads, and Grandparents—based on gender and age, it becomes evident that both Moms and Grandparents are more emotionally susceptible to changes in the family's socioeconomic situation. Interestingly, as Moms and Grandparents age, they tend to report higher levels of happiness. In contrast, the Dads group shows an adverse relationship with age, though this finding is not statistically significant, as depicted in the tables. This disparity might indicate underlying mental health differences between genders.

Additionally, the impact of educational attainment varies across these groups. For fathers, the coefficient for education (0.15915) is not statistically significant, suggesting that the educational level of fathers does not significantly influence their mental health outcomes. This may indicate that other factors, such as occupational or financial stresses, play a more dominant role in shaping fathers' emotional well-being.

For grandparents, higher educational levels are associated with significantly lower levels of negative emotions. The larger magnitude of this coefficient compared to that for mothers implies that education serves as a robust buffer against stressors for older adults, potentially due to enhanced cognitive resources and better problem-solving abilities.

Moreover, the analysis reveals that the Moms group experiences higher mental health problems compared to the other two groups, as evidenced by the significantly higher intercepts (5.48863*** for Income and 5.70546*** for Education). This finding underscores the heightened emotional vulnerability of mothers in response to socioeconomic changes and emphasizes the need for targeted mental health interventions for this demographic.

c) Robustness Check

Data Exclusion

To verify the robustness of the OLS regression model used in this study and to ensure the rigor of the research findings, further analysis was conducted by excluding primary caregivers who are below the legal age of marriage (22 for males and 20 for females) in China. This exclusion was based on the assumption that these younger caregivers may lack the capability to adequately care for their children and to earn a sufficient livelihood. The results of this robustness check are presented in Table 10.

Table 10

	Robust Check	
	log_dass_total	log_dass_total
log_income	-0.06336*** (-3.68)	
internet	-0.46541** (-2.31)	-0.48086** (-2.35)
air_conditioning	-0.09949	-0.10775

	(-1.20)	(-1.31)
doctor	-0.01714	-0.01271
	(-0.20)	(-0.15)
car	0.05085	0.05495
	(0.53)	(0.57)
log_age	-0.48398***	-0.65163***
	(-3.14)	(-3.87)
health	-0.57199***	-0.57462***
	(-6.82)	(-6.82)
c_gender	0.14029*	0.14226*
	(1.79)	(1.82)
c_study	-0.09756*	-0.08156
	(-1.71)	(-1.44)
edu		-0.12647***
		(-3.10)
_cons	5.75313***	5.95610***
	(8.77)	(8.57)
<i>N</i>	907	910
<i>R</i> ²	0.074	0.078

The results show that both the logarithm of household income (log_income) and the educational level of the primary caregivers (edu) remained statistically significant at the 1% level. These findings are consistent with those obtained from the earlier regression analyses, and the robustness checks conducted further substantiate the stability of these results. Consequently, the conclusions of this study are demonstrated to possess considerable robustness, reinforcing the reliability and validity of the research findings.

This robustness check confirms that the relationships between household income, education, and mental health outcomes of caregivers are stable even after excluding younger caregivers who might not fully represent the typical caregiving population due to their age and potential lack of resources. This adds a layer of confidence to the study's implications for policy and intervention strategies aimed at improving caregiver well-being.

Discussion

The analysis revealed several pivotal insights into the relationship between socioeconomic status and the mental health of primary caregivers. Higher household income was consistently associated with lower levels of negative emotions, such as stress, anxiety, and depression. This relationship suggests that financial security significantly enhances emotional well-being by providing better living conditions, improved access to healthcare, and greater opportunities for leisure and recreational activities. The financial stability afforded by higher income levels likely alleviates many of the daily stressors that contribute to negative emotional states.

Educational attainment among primary caregivers also showed a strong negative correlation with negative emotions. Caregivers with higher educational levels reported lower levels of stress, anxiety, and depression, indicating that education equips individuals with essential problem-solving skills, emotional intelligence, and effective coping strategies. These skills are crucial for managing the complex and often demanding responsibilities associated with caregiving. The study also highlighted notable age-related trends. Elderly caregivers exhibited increased sensitivity to changes in socioeconomic status but paradoxically reported higher levels of happiness as they aged. This finding diverges from prior research suggesting that aging correlates with heightened mental health challenges. The increased happiness among older caregivers may be attributed to unique socio-cultural factors and robust community support systems prevalent in rural China.

In contrast, the analysis revealed that mothers face more significant mental health challenges compared to fathers and grandparents. The higher number of stressors experienced by mothers is likely due to their dual roles in caregiving and managing household responsibilities. This underscores the need for tailored mental health interventions that address the specific burdens faced by younger female caregivers.

These findings yield several critical insights. Firstly, supporting the mental health of elderly caregivers through targeted investments in education and healthcare is vital, especially in light of China's aging population. Enhancing educational opportunities and health services can significantly benefit caregivers' mental health, particularly for older adults and mothers. Such investments can bolster caregivers' capacity to manage stress and mitigate negative emotional states.

Future research directions should include investigating the mediating role of parents' mental well-being between socioeconomic status and children's cognitive development. Understanding this relationship can provide deeper insights into how poverty impacts children's long-term outcomes. Additionally, exploring mental health disparities between genders is crucial for developing gender-specific mental health interventions. Incorporating subjective indicators, such as Subjective Social Status (SSS), can also provide a richer understanding of how perceived social positions influence mental health outcomes, offering a more comprehensive view of the psychosocial factors affecting caregivers.

Despite the robust findings, this study has several limitations. The relatively small sample size, confined to a specific region, may limit the generalizability of the results. Future research should aim to include larger, more diverse samples to enhance external validity. The cross-sectional nature of the data restricts the ability to establish causal relationships.

Longitudinal studies are necessary to understand better the dynamics between socioeconomic factors and mental health over time. The small R^2 values suggest that the model does not capture all relevant variables influencing mental health outcomes. Important factors such as occupation, which is not included in the study, and bank savings, which not included in the data, and their inclusion in future research could significantly enhance the model's explanatory power.

Conclusion

This study highlights the significant impact of socioeconomic factors on the mental health outcomes of primary caregivers in rural China. Higher household income and educational levels are associated with lower levels of negative emotions, underscoring the protective effects of financial stability and educational attainment. The results also bring attention to distinct mental health challenges faced by different demographic groups, particularly the heightened vulnerability of elderly caregivers and females. These findings support the need for targeted mental health interventions and policies that address both immediate symptoms and broader socioeconomic determinants. Investing in education and healthcare is essential for improving caregivers' well-being and fostering a supportive environment for early childhood development.

Future research should further explore these relationships by providing a more comprehensive understanding of how socioeconomic status influences mental health over the lifespan. Specifically, investigating the role of parents' mental well-being as a mediator between socioeconomic status (SES) and children's cognitive development could offer deeper insights into how poverty impacts children's long-term well-being. Additionally, exploring the mental health disparities between genders can help understand the unique challenges faced by each group, informing the development of gender-specific mental health interventions. Incorporating subjective indicators such as Subjective Social Status (SSS) would also be valuable in examining the dynamics of perceived social positions and their impact on mental health. By addressing these areas, future research can provide valuable insights to guide policy and programmatic efforts aimed at enhancing the mental health and well-being of caregivers and their children in rural settings.

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